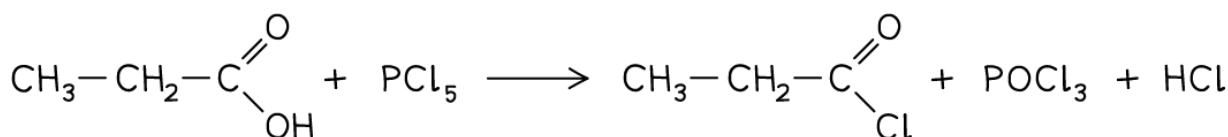


Acyl Chlorides (Cambridge (CIE) A Level Chemistry): Revision Note

Production of Acyl Chlorides

- Due to the increased reactivity of acyl chlorides compared to carboxylic acids, they are often used as **starting compounds** in organic reactions
- Acyl chlorides are compounds that contain an -COCl functional group and can be prepared from the reaction of carboxylic acids with:
 - Solid** phosphorus(V) chloride (PCl_5)
 - Liquid** phosphorus(III) chloride (PCl_3) and heat
 - Liquid** sulfur dichloride oxide (SOCl_2)
- Propanoyl chloride can this way be prepared from propanoic acid using the reactions above

Using propanoic acid to form propanoyl chloride



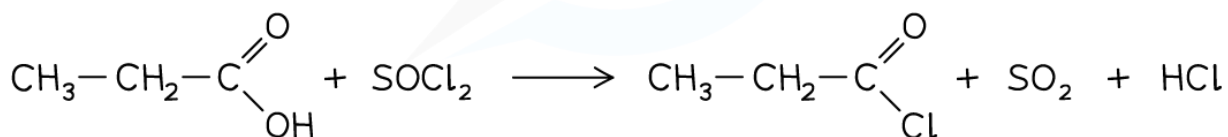
PROPANOIC ACID

PROPANOYL CHLORIDE



PROPANOIC ACID

PROPANOYL CHLORIDE



PROPANOIC ACID

PROPANOYL CHLORIDE

Propanoic acid can be used to produce propanoyl chloride with different by-products depending on the reagent used

Reactions of Acyl Chlorides

- **Acyl chlorides** are **reactive** organic compounds that undergo many reactions such as **addition-elimination reactions**
- In addition-elimination reactions, the **addition** of a small molecule across the C=O bond takes place followed by **elimination** of a small molecule
- Examples of these addition-elimination reactions include:
 - **Hydrolysis**
 - Reaction with alcohols and phenols to form **esters**
 - Reaction with ammonia and amines to form **amides**

Hydrolysis

- The **hydrolysis** of acyl chlorides results in the formation of a **carboxylic acid** and **HCl** molecule
- This is an **addition-elimination** reaction
 - A **water molecule** adds across the C=O bond
 - A hydrochloric acid (HCl) molecule is **eliminated**
- An example is the hydrolysis of propanoyl chloride to form propanoic acid and HCl

Hydrolysis of acyl chlorides

Acyl chlorides are hydrolysed to carboxylic acids

Formation of esters

- Acyl chlorides can react with **alcohols** and **phenols** to form esters
 - The reaction with phenols requires **heat** and a **base**

- Esters can also be formed from the reaction of **carboxylic acids** with phenol and alcohols however, this is a **slower** reaction as carboxylic acids are less reactive and the reaction does **not go to completion** (so less product is formed)
- Acyl chlorides are therefore more useful in the synthesis of esters
- The esterification of acyl chlorides is also an **addition-elimination** reaction
 - The alcohol or phenol adds across the C=O bond
 - A HCl molecule is eliminated

Esterification reactions using acyl chlorides

Acyl chlorides undergo esterification with alcohols and phenols to form esters

Formation of amides

- Acyl chlorides react with **ammonia** or **primary amines** to form **amides** in a **condensation reaction**.
- A **lone pair** on the nitrogen atom attacks the **carbonyl carbon** in the acyl chloride.
- The reaction proceeds via a nucleophilic **addition-elimination mechanism**:
 - The nucleophile adds to the C=O bond
 - A chloride ion (Cl⁻) is eliminated
 - **Hydrogen chloride (HCl)** is formed

What happens to the HCl?

- The **HCl formed** does not remain unreacted.

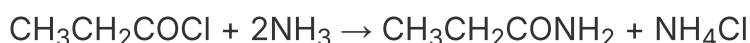
- It is **immediately neutralised** by a **second molecule** of ammonia or amine present in excess.
- This forms an **ammonium salt** (e.g. NH_4Cl , $\text{CH}_3\text{NH}_3\text{Cl}$).

Why 2 molecules are needed

- The **1st molecule** of ammonia/amine forms the **amide**
- The **2nd molecule** of ammonia/amine neutralises the **HCl** and forms **ammonium salt**

Examples

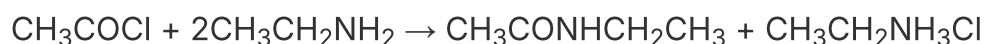
- **Reaction with ammonia**
- **Product:** Primary amide (propanamide) and ammonium chloride



- **Reaction with methylamine**
- **Product:** Secondary (substituted) amide and methylammonium chloride



- **Reaction with ethylamine**
- **Product:** Secondary (substituted) amide and ethylammonium chloride



Summary for formation of amides

- All reactions form **HCl**, which is **not observed** as a separate product
- The **HCl is neutralised** by excess ammonia or amine
- The final products are an **amide** and an **ammonium salt**